Compute the PageRank scores on the given dataset

- Dataset: Data.txt
- The format of the lines in the file is as follow:

  FromNodeID ToNodeID
- In this project, you need to report the **Top 100 NodeID** with their PageRank scores. You can choose different parameters, such as the teleport parameter, to compare different results.
- One result you must report is that when setting the teleport parameter to 0.85.
- In addition to the basic PageRank algorithm, you need **to optimize your memory use** via Block Matrix, Sparse Matrix and some other approches.
- Deadline: 2026.5.15

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Specific Requirement:

- Language: You can use C/C++/Python.
- For C/C++, It's recommended to code in a single file and minimize the use of third-party libraries. You'd better to use gcc or g++ for compiling your code and give the compilation parameters in *compile-parameter.txt* with the following format.

  g++ -fopenmp PageRank.cpp -o PageRank.exe

- For Python, It's recommended to use commonly-used libraries or package (e.g. numpy and scipy is accepted). To confirm the availability of a package, you can email *bigdatacomputing@163.com* . It's recommended to code in a single file. Ensure the entry of your code should be named as 'main.py'.

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Specific Requirement:

- Consider of dead-ends and spider-traps
- Optimize your memory use **as much as possible.**
- You can optimize the memory use in other ways**, but Block Matrix and Sparse Matrix(https://zhuanlan.zhihu.com/p/557231877) optimization is compulsorily required**
- Your program is required to iterate until reaching convergence
- Using existing PageRank API is prohibited. (e.g. *networkx.pagerank* in Python)
- Your program need to read *Data.txt*, conduct PageRank and Return the Top-100 nodes in *Res.txt* with its score as following format:

  NodeID Score

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Specific Requirement:

- You'd better make sure the maximum memory use during the whole life of your program should be lower than **80MB.**
- You can't to sacrifice too much time performance to minimize your memory usage. The program need to complete its runtime under **60s**.
- Maybe you can use some Parallel or/and Distributed Computing techniques. **But it's not the main content of this assignment**.

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Submitting:

- *bigdatacomputing@163.com*
- *Report:* Include but not limited: description of dataset, key code details, **how to optimize the memory usage** and result with analysis
- *Code:* Meet the above requirements, Input Dataset is not required
- *Result:* Named as *Res.txt* and Meet the above requirements with a teleport parameter of 0.85
- *Executable File:* Compile with all **Static link library**.
  - For C/C++, maybe you can use [-*static*] setting in gcc/g++
  - For Python, using third-party packages to generate and integrate the package used in your code.
  - **Ensure your code can run in other computers (and mainstream OS)**. (You can ask LLMs or TAs for more).

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Submitting:

- Indicate team division of work and contributions.
- If no team division indicated, whole group will be given the same score.
- Compress your submission (zip format), with following naming:

  2000000_小张_2111111_小王_2222222_小林_第一次作业.zip
- Deadline: 12:00(UTC+8), 2026.5.15
- 10% of the points will be deducted for each day of late submission.
- **Submitting just one or two days before DDL is not recommended.**

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Hints:

- You **WILL NOT** get higher score only because you complete it solely (we recommend to complete it as a teamwork).
- You **CAN** use LLM for this work, **but purely AI-gen Report is explicitly prohibited.**
- **Failure to comply with the format will result in failure to grade.**
- **Code plagiarism and academic fraud are absolutely intolerable.**

Supplementary Materials

南开大学教务部文件 教字〔2025〕5 号

南开大学本科教学人工智能工具使用规范（试行）

第八条 允许使用的范围。在征得任课教师或指导教师同意的前提下，且 AI 工具辅助完成的内容不会对课程作业、毕业论文（设计）、实验报告等的核心内容和创新能力考查造成影响时，学生可以在以下范围内合理使用 AI 工具。

1.允许使用 AI 工具进行文献检索、资料收集、关键词推荐和文献管理，但须确保所引用文献的真实性和可靠性。

2.允许使用 AI 工具进行语法检查、拼写校正、格式排版等基础性、非创造性工作，但需标注生成内容并限制比例。语言类学科专业应对使用标准另行明确限定。

3.允许使用 AI 工具进行基于真实数据的图表制作、辅助制图，但应确保最终图表的原创性。

4.允许使用 AI 工具进行程序代码的辅助编写、调试和纠错，所有代码或算法须经过作者审核和测试。

第九条 禁止使用的范围。

1.禁止使用 AI 工具进行任何涉及敏感信息（如个人隐私、商业秘密等）或涉及保密的内容创作，禁止将涉密内容上传至任何 AI 平台。

2.禁止使用 AI 工具生成或篡改原始核心论点、实验数据、调研数据，禁止使用 AI 工具生成基于虚假研究数据的图表、图片。

3.禁止使用 AI 工具进行代写、抄袭、剽窃等学术不端行为。

4.禁止在以上机考试、结课论文、笔试等形式开展的期中、期末测试中使用 AI 工具。特殊情况确需使用的，任课教师应提前予以说明。

5.禁止使用 AI 工具直接替代任课教师或指导教师进行指导、评审、评价等教学工作，以确保学习效果评价的客观性和学术严谨性。